1 Inputs

Global land · 0.25° annual 1982–2021

Coherent precipitation (P) and evaporation (E) candidate pairs

1	ERA5-Land	ERA5-Land
2	FLDAS	FLDAS
3	MERRA-2	MERRA-2
4	TerraClimate	TerraClimate
5	MSWEP	GLEAM

Reference ensembles

8 P | 8 E

For evaluation only

Potential evaporation (PET)

12 **PET estimates**

For plausibility screening only

2 Candidate-specific evaluation

Per grid cell for P and E

Calculate statistics

Mean
Standard deviation (sd)
Sen-slope
Trend significance

Leave-candidate-out



Median of means
Median of sd
Median of Sen-slope
Majority significance

Calculate 4 error metrics

Relative absolute bias:
- mean
- sd
- Sen-slope

Significance agreement

3 Sampling probabilities

Use 4 **error metrics** to calculate **performance scores** and **ranks**

Grid-cell weights

10 scenarios Table X
***2 product scenarios**:*
Base and Change
8 sensitivity scenarios

Physics gate

For candidate pairs, is **P/E ≥ 0.9**?
AND
For at least 7 of 12 **PET** estimates, is **E < PET**?
No ➔ Pair's grid-cell weight = 0

Sampling probabilities

For each scenario and **candidate pair,** calculate area-weighted **sampling probabilities** for *235* **Biome x IPCC region** *sampling units*

4 Monte Carlo sampling

For all 235 units biome x IPCC region

For **100 members,** sample a uniform random number in [0,1]

Reuse that number across all **10 scenarios**

Select candidate P–E pair using unit's cumulative scenario sampling probabilities

Use the selected pair's original **P** and **E** values for all cells and years within the **unit**

➔ For the same **member,** differences in **P** and **E** values come from **scenario** weights

Aggregation

10x100 area-weighted annual **P** and **E** time series for each **IPCC region** and **globally**

5 ITHACA products

Variables
P and E

Scale
Grid
Biome x IPCC region
IPCC region
global

Records
Monte Carlo members
Scenario weights
Candidate pair provenance
Uncertainty estimates

Two probabilistic **P–E** ensemble products of plausible terrestrial **water-cycle worlds.**
Base: balanced weighting
Change: greater weight on trends